

elliott.roberts@maine.edu • [linkedin.com/in/elliottroberts/](https://www.linkedin.com/in/elliottroberts/) • elliottroberts.github.io • Beals, ME, U.S.A.

Oceanographer with 15+ years’ experience in chemical oceanography, climatology, and marine biogeochemistry field research. Published author with four peer-reviewed publications, one first-author publication in progress, and presentations in France, China, Singapore, and U.S.A. Excellent writing and communication skills. Capable of using new and existing data in novel research to investigate a broad range of topics while controlling data quality and documenting progress. Technical proficiency using Python, R, MATLAB, SQL, Surfer, Ocean Data View, IBM Watson Studio, and Microsoft Office Professional Suite.

Areas of Expertise include:

Field Research	Project Management	Relationship Management	Data Interpretation
Oceanography	Protocol Design	Team Leadership	Machine Learning
Climatology	Data Collection & Analysis	Research Collaboration	Computation Analysis
Carbon Cycle	Publications & Presentations	Public Relations	Model Development
Marine Biology	Database Management	Training & Mentorships	Model Evaluation

Professional and Research Experience

Downeast Institute • Beals, Maine, U.S.A. • May 2025 to Present

Research Scientist – Ocean Acidification and Coastal Resilience

- Design and conduct lab and field experiments on the effects of carbonate chemistry and warming on shellfish life-history dynamics.
- Maintain, operate, and improve the Ocean Acidification laboratory, including factorial tank systems, and manage procurement of equipment and supplies for research support.
- Apply expertise in shellfish biology and carbonate chemistry to experimental design and data interpretation.
- Write grant proposals and collaborate on externally funded research projects.

NuWa Solutions • Dover, Delaware, U.S.A. • March 2025 to Present

Founder

Founded a climate-tech company focused on ocean-based carbon dioxide removal, primarily via ocean alkalinity enhancement (OAE). Current non-high-level activities so far:

- Coordinated with government-affiliated scientists across multiple coastal regions to evaluate site feasibility and research alignment for future OAE field trials
- Submitted two Carbon-to-Sea RFI proposals for field sites where NuWa would be the CDR developer – Maine & Costa Rica
- Advised early-stage mCDR startups on research strategy, stakeholder engagement, and MRV framework development
- Completed the NSF I-Corps Spark program to apply customer discovery and market validation tools to early-stage climate tech development

Ephemeral Carbon • Santa Clara, California, U.S.A. • June 2024 to September 2024

Lead Oceanographer

- Developed innovative satellite-based techniques for selecting field sites, integrating big data on sediment types to enhance project outcomes.
- Contributed to ongoing Monitoring, Reporting, and Verification (MRV) frameworks to ensure transparent and effective carbon capture practices.

- Contributed to the development of methodologies for carbon sequestration, focusing on chemical and sediment-based approaches.
- Gained extensive experience in international business negotiations, securing key partnerships for large-scale environmental projects.
- Played a pivotal role to securing V05 Voucher funding from the U.S. Department of Energy by contributing to the development of a novel approach to ocean alkalinity enhancement for scalable carbon sequestration.

University of Delaware • Newark, Delaware, U.S.A. • July 2021 to December 2023

Postdoctoral Research Scientist

Worked on estimating the status of the surface ocean biological pump/marine productivity and terrestrial and ocean margin carbon export fluxes in the Gulf of Mexico and US eastern ocean margins as well as assisting in estuarine research. Contributed to NOAA-funded research projects at the Cai Lab at the University of Delaware such as organizing GOMECC-4, ECOA-3, and GO-SHIP expedition preparations for $\delta^{13}\text{C}$ -DIC and O_2/Ar measurements. Built the website of the Cai lab.

- Main researcher for surface underway O_2/Ar determinations for the GOMECC-4 and ECOA-3 expeditions to establish the surface net community production status of the Gulf of Mexico and the North American East Coast, respectively. Assisted in collecting and analyzing 2019 $\delta^{13}\text{C}$ -DIC samples from GOMECC-4 and 2148 TA (total alkalinity) samples from ECOA-3.
- Main researcher for surface underway $p\text{CO}_2$ determinations in the Chesapeake Bay plume region to determine the seasonal $p\text{CO}_2$ surface variability in this area. This involved the collection and analysis of discrete $\delta^{13}\text{C}$ -DIC, TA, pH, and salinity samples.
- Assisted with the first-ever large-scale evaluation of fast-responding pH sensors built around the Honeywell DuraFet over a wide range of salinity between 1 and 30 - determining how accurate these widely used and commercially-available pH sensors are in dynamic nearshore environments
- Measured TA and total dissolved inorganic carbon (DIC) on discrete water samples collected as part of a pilot study of estuarine and coastal ocean acidification in Delaware's Inland Bays.

Xiamen University • Xiamen, China • March 2019 to March 2020

Assistant Research Scientist

Integral member of many State Key Laboratory of Marine Environmental Science research projects. Collaborated with an international science team to complete critical research; analyzed data, authored and co-authored major publications, and wrote and implemented the sampling protocol for CFCs and SF_6 . These results were presented at international conferences. Calculated man-made carbon contribution to the South China Sea, determined the amount of carbon transferring from the North Pacific Ocean to the South China Sea, and outlined seasonal fluctuations of carbon in the South China Sea.

- Supervised four student researchers to collect and analyze 144 CFC and SF_6 samples; applied parameters and 7120 samples from the CO_2 -carbonate and nutrient system to determine man-made carbon concentrations.
- Led research that discovered South China Sea deep water (>2000m) had higher average carbon ($23.68 \mu\text{mol kg}^{-1}$ difference) than the West Philippine Sea (North Pacific Ocean) due to more organic matter decomposition.
- Detected average Kuroshio contribution to DIC inventory in the upper 150 m of the ocean was seasonally significant, ranging between 11% and 32% with the highest contributions found during spring and winter.
- Discovered that between 500 and 1500m, $1.0 \pm 0.4 \text{ Tmol yr}^{-1}$ of man-made carbon is transported from the South China Sea to the West Philippine Sea. This suggests that marginal seas could play a considerable role in contributing man-made carbon into nearby coastal systems and influencing the marine carbon budget in those regions.
- Data taken during 2009, 2011, 2014, and 2016 showed a substantially high increase in man-made carbon (~ 22.5 - $27.7 \mu\text{mol kg}^{-1}$) within surface water, which would result in significant increases in ocean acidification that would have immense effects on marine life.

The Scripps Research Institute • San Diego, CA, U.S.A. • June 2010 to September 2010

Database Assistant

Supported chemists by entering biological data into database. Input molecular compound information into a database; maintained the database by updating software.

- Supported 12 chemists during data collection by submitting 10+K data entries into a database.

Selected Additional Experience

Staff Associate, Lamont-Doherty Earth Observatory, Columbia University, NY, U.S.A. (2016)

Visiting Scientist, Scripps Institute of Oceanography, University of California, San Diego, CA, U.S.A. (2015)

Education

Doctor of Philosophy in Marine Chemistry, (September 2013 – June 2019; 6 years)

Xiamen University • Xiamen, China • GPA: 4.0 • Sino-US Fellowship

Thesis: Carbonate Chemistry and Anthropogenic CO₂ in the northern South China Sea Basin

Courses:

Coastal Sea Chemistry (近海化学)	Standard Score: 100%
Marine Biogeochemical Cycles (海洋生物地球化学循)	Standard Score: 100%
Marine Biogeochemistry (海洋生物地球化学)	Standard Score: 100%
Marine Paleoceanography & Climatology (古海洋学与古气候学)	Standard Score: 95%

Master of Research in Ocean Science (2011)

University of Southampton • Southampton, UK

Bachelor of Science in Marine Biology (2010)

Hawaii Pacific University • Honolulu, HI

Selected Certificates & Licenses

IBM Data Science Specialization, Certificate with Distinction, IBM via Coursera (2020)

Machine Learning using Python, Certificate with Distinction, IBM via Coursera (2020)

Databases and SQL for Data Science, Certificate with Distinction, IBM via Coursera (2020)

R Programming, Certificate with Distinction, John Hopkins University via Coursera (2015)

Training & Development

SOLAS Workshop, Corsica, France (2018)

- Understood oceanic, atmospheric, and air-sea interface biogeochemistry with physics, and the relationships within them.
- Studied geoengineering, climate change and variability, interpreting atmospheric and oceanic models.
- Proficient in directly measuring and calculating air-sea fluxes using eddy covariance, operating the PISCES biogeochemical model, and utilizing remote sensing data.
- Press, and social media communication training.

Metal Speciation and Isotopes in the Ocean, GEOTRACES and Beyond, Xiamen University (2018)

Knowledgeable about metal marine biogeochemistry, metal and barium isotopes in the water column and marine sediment, metal speciation in the ocean, trace metal-phytoplankton interactions, trace metal bioavailability in an acidifying ocean, transformation and fluxes of micronutrient metals across seafloor redox interfaces and its processes and approaches, and the application of radio-isotopes in biological pump studies

- Understood how to sample oceanic trace elements: use a Nu Plasma MC-ICP-MS, conduct a flow injection analysis, use a clean van.
- Proficient on electrochemistry and metal speciation methods: how to create solid state electrodes, determine redox species and their isotopes.
- Emphasized on metal-ligand solution complex interactions with planktonic cell surfaces.

Satellite Oceanography, Second Institute of Oceanography (2017)

- Created satellite orbits to suit measurement and instrument design.
- Comfortable with mechanisms associated of IR measurements (SST), Ocean Color, Passive Microwaves, Altimetry, Scatterometers, terrestrial sensor instruments (e.g. LANDSAT), and SAR; including how to process and analyze samples from received sensor data.
- How to composite (smooth) and spatially filter images produced from received data.

Climate Mathematics, Scripps Institute of Oceanography (2015)

- Emphasis on Calculus, Statistics, Matrices, Multivariate Regression, Energy Balance Models for Climate, Differential Equations, and Conservation Laws in Climate Dynamics
- Trained in R-Programming
- *Project Title: 15-year Evolution of Anthropogenic Carbon in the Caribbean Sea using a Back-Calculation Method*

Professional & Community Involvement

Member, Royal Society of Biology (2018 - Present)

Associate Member, Royal Society of Chemistry (2016 - Present)

Member, American Geophysical Union (2014 - Present)

Member, Association for the Sciences of Limnology and Oceanography (2014 - Present)

President's Host, Hawaii Pacific University (2010)

Programming Experience

Language • Python, R, Matlab, SQL, Apache Spark, Excel

IDEs and Visualization Software • Spyder, RStudio, Surfer, Grapher, Voxler, Ocean Data View, Zeppelin

Publications

Peer-Reviewed Publications

Roberts, E. G., Z. Ouyang, Q. Li, L. Barbero, D. Pierrot, C. Featherstone, C. Langdon, A. Stefanick, W-J Cai (2025) Summertime net community production (NCP) via underway measurements of O₂/Ar and its control on CO₂ flux in the northern Gulf of Mexico. *The Science of the Total Environment*

Wang, Z., Z. Cao, Z. Liu, W. Zhai, Y. Luo, Y. Lin, **E. Roberts**, J. Gan, M. Dai (2024). Pacific Ocean originated anthropogenic carbon in the South China Sea and their impact on ocean acidification. *Science Advances*.

Meilana, L., Q. Fang, **E. G. Roberts**, H. A. Susanto, H. O. Ikhumhen, B. R. Nelson (2024). Identification of potential marine protected areas based on cumulative risk approach: informed marine spatial planning. *Ocean and Coastal Management*.

- S. F. Gonski, G. W. Luther, III, A. L. Kelley, T. R. Martz, **E. G. Roberts**, X. Li, B. Dong, J. A. Watson, Y. Takeshita, T. S. Wirth, N. Hussain, R. J. Feris Serrano, E. Hale, W-J. Cai. (2024). A Half-Cell Reaction Approach for pH Calculation using a Solid-State Chloride Ion-Selective Electrode and the Honeywell Durafet. *Marine Chemistry*,
- Roberts, E. G.**, M. Dai, Z. Cao, W. Zhai, L. Guo, Y. Li, S.S.P. Shen, C. Du. (2021). The Carbonate System of the Northern South China Sea: Seasonality and Exchange with the Western North Pacific. *Progress in Oceanography*, doi: 10.1016/j.pocean.2020.102464
- Du, C., J. Gan, C. R. Hui, Z. Lu, X. Zhao, **E. Roberts**, M. Dai (2020). Dynamics of Dissolved Inorganic Carbon in the South China Sea: A Modelling Study. *Progress in Oceanography*, doi: 10.1016/j.pocean.2020.102367
- Liu Q., X. Guo, Z. Yin, K. Zhou, **E. G. Roberts**, M. Dai (2018). Carbon Fluxes in the China Seas: An Overview and Perspective. *Science China Earth Sciences*, 61, doi: 10.1007/s11430-017-9267-4.
- Lin, H., M. Dai, S.-J. Kao, L. Wang, **E. Roberts**, J.-Y. T. Yang, T. Huang, and B. He (2016). Spatiotemporal Variability of Nitrous Oxide in a Large Eutrophic Estuarine System: The Pearl River Estuary, China. *Marine Chemistry*, 182, 14-24, doi:10.1016/j.marchem.2016.03.005.

In-Prep Publications

- Roberts, E. G., et al.**, (In Progress). Surface seasonal variability of the marine carbonate system along the Delmarva Peninsula and the Chesapeake Bay Plume region. Manuscript in preparation.
- Roberts, E. G.**, et al., (In Progress). Surface CO₂ Fluxes and Marine Productivity: Estimations Along the North American East Coast Using O₂/Ar-Based Net Community Production (NCP). *Science Advances*
- Roberts, E. G.**, W-J Cai, Z. Ouyang, N.Hussain, D. Pierrot, C. Featherstone, Z. Wu, (In Prep.). Oceanic O₂/Ar-Based Estimates of Net Community Production (NCP): Modulating Summertime Surface CO₂ Fluxes and Marine Productivity in the northwestern Atlantic Coast. *The Science of the Total Environment*
- Roberts, E. G.**, Q. Li, W-J Cai, Z. Ouyang, L. Barbero, C. Featherstone, C. Langdon, Z. Wu, A. Stefanick, W-J Cai (In Prep.). High-Resolution Insights into Summertime Surface Marine Productivity and CO₂ flux Patterns in the Gulf of Mexico. *Science Advances*
- Roberts, E. G.**, M. Dai, S. S. P. Shen, Y. Zhu, X. Guo, W. M. Smethie Jr., E. Gorman, C. Addey, L. Guo, L. Wang, Y. Xu, C. Du (Near Submission). Anthropogenic Carbon Penetration and Distribution in the Northern South China Sea Basin. *Nature Climate Change*
- Roberts, E. G.**, M. Dai, X. Guo, C. Addey L. Guo, L. Wang, Y. Xu, W. M. Smethie Jr., E. Gorman, S. S. P. Shen, Y. Zhu (Near Submission). Carbon Cycling in China Seas (CHOICE-C) II: High-Resolution Insights on Carbonate System Dynamics in the South China Sea. *Earth System Science Data*

Field Experience

- NOAA (2023)** • *R/V Sharp* • Chesapeake Bay Plume Cruise • 21 days; 3 cruises total • Underway pCO₂, discrete ¹³C-DIC, TA, pH, and salinity • University of Delaware
- NOAA (2022)** • *R/V Ronald H. Brown* • East Coast Ocean Acidification (ECO-A3) Cruise • 50 days total • underway O₂/Ar (net community production), underway DO, and TA • University of Delaware
- NOAA (2021)** • *R/V Ronald H. Brown* • Gulf of Mexico Ecosystems and Carbon (GOMECC-4) Cruise • 39 days total • underway O₂/Ar (net community production), underway DO, and discrete ¹³C-DIC • University of Delaware
- SIO & WHOI/MIT (2012)** • *R/V New Horizon* • NH1208 – Ocean Acidification Pteropod Study Cruise • 20 days total • discrete spec pH • Woods Hole Oceanographic Institution
- GEOTRACES (2011)** • *R/V Discovery* • D361 – GEOTRACES Cruise • 20 days total • DIC, TA, pH • University of Southampton / National Oceanography Centre, Southampton

Data Products & Open Datasets

- Roberts, Elliott G.; Ouyang, Zhangxian; Hussain, Najid; Cai, Wei-Jun (2023). Biological oxygen saturation ($\Delta(O_2/Ar)$) collected from surface underway observations using EIMS from NOAA Ship Ronald H. Brown during the ECOA2022 project from 2022-08-06 to 2022-09-23 (NCEI Accession 0281870). NOAA National Centers for Environmental Information. Dataset. <https://www.ncei.noaa.gov/archive/accession/0281870>.
- Roberts, Elliott G.; Cai, Wei-Jun; Hussain, Najid (2023). Partial pressure of CO₂ (pCO₂), water temperature and salinity collected from surface underway observations from R/V Hugh R. Sharp in the Delmarva Peninsula and Chesapeake Bay Plume Region, North Atlantic Ocean from 2023-04-10 to 2023-04-14 (NCEI Accession 0285104). NOAA National Centers for Environmental Information. Dataset. <https://doi.org/10.25921/j2pg-4t50>.
- Roberts, Elliott G.; Li, Qian; Cai, Wei-Jun (2023). Biological oxygen saturation ($\Delta(O_2/Ar)$) data collected from surface underway observations using EIMS during the NOAA Ship Ronald H. Brown GOMECC2021 project cruise in the Gulf of Mexico from 2021-09-14 to 2021-10-20 (NCEI Accession 0282979). [indicate subset used]. NOAA National Centers for Environmental Information. Dataset. <https://doi.org/10.25921/dcw8-d577>.

Teaching Experience

University of Bamenda • Bamenda, Cameroon • 2020 – Present

Guest Lecturer (Remote) – On Call

Lectured University of Bamenda Environmental Science undergraduates, who are affected by the Ambazonia War, on climatology and oceanography

- Old lecture material can be found on the website: elliottroberts.github.io/

Xiamen University • Xiamen, China • September 2013 to June 2017

Scientific English Instructor

Facilitated technical and scientific English classes for Ocean Carbon Group researchers.

- Mentored 30 scientists and technicians and taught two classes weekly.

Oxford Tutoring • Oceanside, CA, U.S.A. • October 2011 to June 2013

Instructor

Taught socioeconomically disadvantaged students in all courses at Oceanside High School.

- Tutored 40 failing students each semester and achieved 100% passing rate.

Selected Participated Conferences

Carbon-to-Sea Initiative Annual Convening • Washington D.C., U.S.A. (2025)

Ocean Visions Summit • Vancouver, Canada (2025)

NOAA Ocean Acidification Program (OAP) Community Meeting • San Diego, U.S.A. (2023)

National Major Scientific Research Program (973) Meeting – South China Sea carbon cycle process, mechanism and its global significance: 2nd Round • Xiamen, China (2019)

National Major Scientific Research Program (973) Meeting – South China Sea carbon cycle process, mechanism and its global significance • Tianjin, China (2019)

3rd Xiamen Symposium on Marine Environmental Sciences (XMAS) • Xiamen, China (2017)

Asia Oceania Geochemical Sciences (AOGS) • Singapore (2015)

2nd Xiamen Symposium on Marine Environmental Sciences (XMAS) • Xiamen, China (2015)

2nd Seafloor Observation Symposium • Xiamen, China (2014)
CHOICE-C I Assessment and CHOICE-C II Commencement • Xiamen, China (2014)
Analytica China • Shanghai, China (2014)
1st Xiamen Symposium on Marine Environmental Sciences (XMAS) • Xiamen, China (2014)

Selected Media Mentions

“Threat 3 – Estuarine Nutrient Sources and Loading, Eutrophication, and Acidification Lead” • Lewes, Delaware, U.S.A. (December 2022)

Interview • Published on X / Twitter • **Project WiCCED**

- Discussed my background, recent work, and my thoughts on the environment.

“How do we measure productivity in the ocean?” • Newark, Delaware, U.S.A. (August 2022)

Interview • Published on X / Twitter, Facebook, and Instagram • **NOAA Ocean Acidification Program (OAP)**

- Discussed my role in an upcoming NOAA-led expedition at the time (East Coast Ocean Acidification – 3)

References available upon reasonable request